

BUIDL CTC 2026 FALL · DEFI · RWA

ProofLine

Verified economic history becomes reusable credit.

Ethereum contains evidence of economic behaviour. Attestcoin lets Creditcoin verify that evidence. ProofLine turns verified events into persistent credit state that independent applications can consume.

Live on Ethereum Sepolia & Creditcoin CC3 · 118 tests · every claim reproducible from the repo

THE PROBLEM

Economic behaviour is provable. Creditworthiness is not portable.

A business earns on Ethereum. That history is public, permanent and cryptographically real — and completely unusable as credit anywhere else.

Bridging it means trusting an operator. Re-stating it means trusting a database. Both replace the one property that made the history worth having.

The question is not how to move the data. It is how to make a credit decision that is impossible to make without the proof.

Remove the proof layer from ProofLine and the credit file cannot change at all

Attestcoin proves inclusion. It does not prove truth.

A proof establishes that an authorised contract emitted a log in a transaction that succeeded. It says nothing about whether the log's claim is *true*.

So the source contract has to make a false claim unreachable. `InvoicePaid` cannot exist unless the measured ERC-20 balance delta equals the invoice amount. `InvoiceLate` cannot exist before the due date.

Attestcoin then proves the contract reached a state its own invariants made impossible to reach falsely.

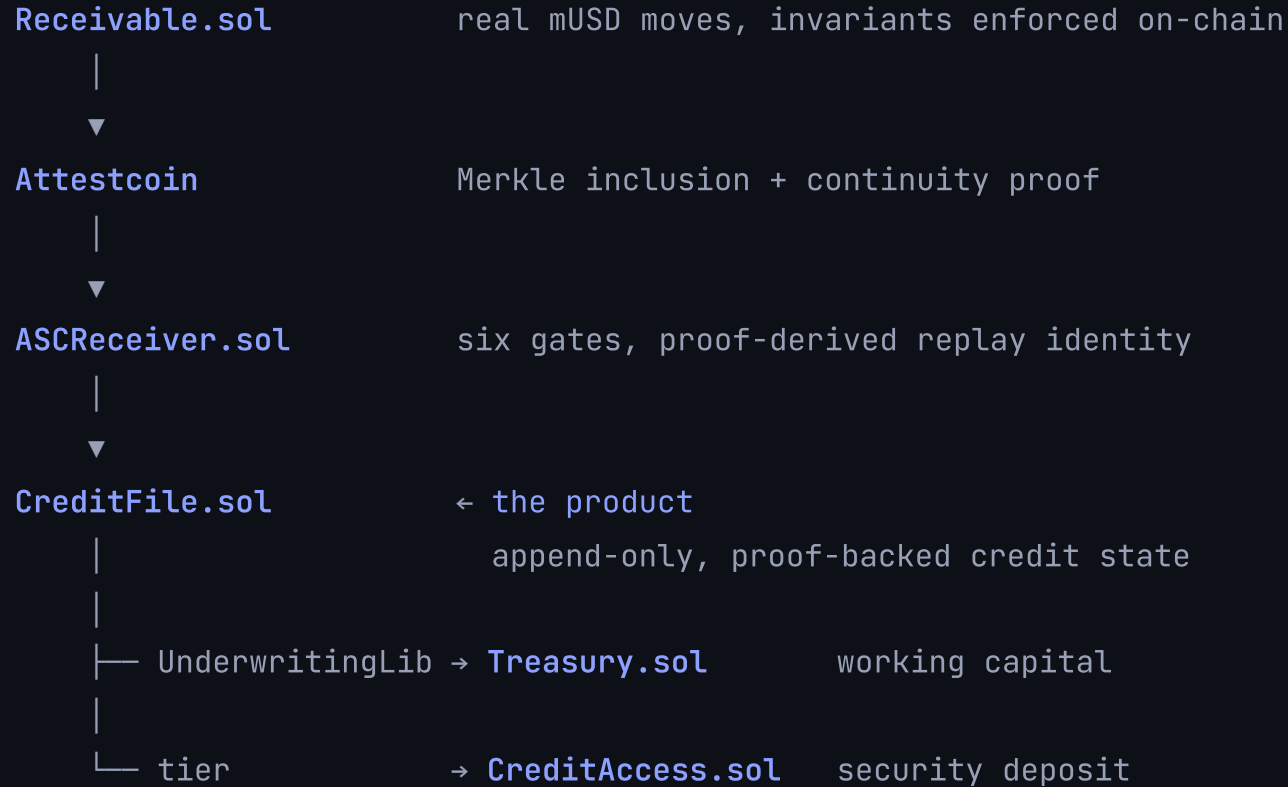
SIX VERIFICATION GATES

- 1 replay proof-derived identity
- 2 verifyAndEmit precompile
- 3 transaction type
- 4 `receiptStatus = 1` protocol MUST
- 5 matching logs found
- 6 emitter authorised

Gate 6 must come last: the emitter address does not exist until the receipt is decoded, which requires the proof to have verified.

HOW IT WORKS

Ethereum Sepolia



One writer. CreditFile changes only when a proof verifies, in the same transaction

Five settlements changed the terms

SETTLEMENT	COUNTERPARTY	AMOUNT	TIER	ADVANCE	APR	DEPOSIT
#1	buyer A	\$12,000	STANDARD	60%	16%	75%
#2	buyer B	\$8,000	STANDARD	60%	16%	75%
#3	buyer C	\$10,000	GOOD	70%	14%	40%
#4	buyer A	\$7,000	GOOD	70%	14%	40%
#5	buyer B	\$9,000	TRUSTED	80%	12%	0%

Every row is a real Sepolia transaction moving real value between independently registered counterparties, proven through Attestcoin. Nothing was seeded. **Capacity rose from 7,200 to 9,600 while the largest settlement stayed at 12,000** — every increase came from a rate earned by history, never from a bigger invoice.

THEN WE ACTUALLY LENT

Earned credit, used

STATE	TIER	CAPACITY	APPROVED LINE	AVAILABLE	DEBT
Before receivable	TRUSTED	9,600	9,600	0	0
Receivable outstanding	TRUSTED	9,600	9,600	9,600	0
Borrowed 6,300	TRUSTED	9,600	9,600	3,300	6,300
Repaid 3,150	TRUSTED	9,600	9,600	6,449.999	3,150.001

The first row is the architecture in one line. Earned standing of 9,600 with nothing currently drawable — a state that cannot be expressed at all if the three numbers are collapsed into one "credit limit".

Tier, capacity and line never moved. Only availability and debt responded to borrowing. The .001 is real accrued interest, left unrounded.

borrow 0xd3174e51 · repay 0x3cf490de · ten assertions, all passing

WHY THIS IS DIFFERENT

CreditFile is the product

REUSABLE

2

Independent consumers over one credit file. Treasury lends; CreditAccess waives security deposits. Neither imports the other, neither computes a tier, neither can write.

LOAD-BEARING

1

One writer. Delete Attestcoin and the credit file is immutable — not degraded, not stale. The financial state transition becomes impossible.

STATEFUL

10

Credit is persistent state, not a one-shot reaction to a proof. History compounds; delinquency tightens terms immediately; a proven default freezes permanently.

One proven settlement moved Treasury from 70%/14% to 80%/12% and waived CreditAccess's 40% deposit — through a credit file that imports neither application. **The invoice contract is our source of evidence, not the product.**

Any Creditcoin application can read `getCreditFile` / `getCreditEvents` / `getTerms`

EVIDENCE & LIMITATIONS

MEASURED, NOT ASSUMED

- **7.96–8.7 min** end-to-end attestation
- **321,498 gas** mean production ingest, 8 samples
- **244–669 ms** proof construction once attested
- **Permissionless relay works** — the worker is convenience, not trust
- Continuity roots vary **1–10** across six blocks — the expiry mechanism

FOUND, FIXED, RECORDED

- Replay key was **caller-spoofable**
- Accounting silently **aliased its input**
- A gate-ordering test **passed while proving nothing**
- Unchecked payer allowed **manufactured counterparties**
- One of our own claims was an inference — **disproved and corrected in place**

STATED PLAINLY

- **Sybil resistance is not solved.** Five controls raise the cost; three funded addresses can still manufacture history. Production needs counterparty attestations and stake-at-risk.
- **Repayment is unsecured.** Writability is in audit, so the receivable cannot be seized. Enforcement is the credit file — deliberate, and Creditcoin's own thesis.
- **Batch proving is unmeasured.** Nothing depends on it.
- **Tier thresholds are our policy**, not a Creditcoin standard. The reusable part is the shape: verified history → deterministic function → terms.

A credit primitive other Creditcoin applications build on — with two of them already shipped, and every claim checkable with `cast call`.